

PROJECT DOCUMENTATION

OptiCrop: Smart Agricultural Production Optimization Engine

Name: swapna konduri

Email: swapnakonduri9@gmail.com

Domain: Artificial Intelligence & Machine Learning

Project Type: Machine Learning with Flask Web Application

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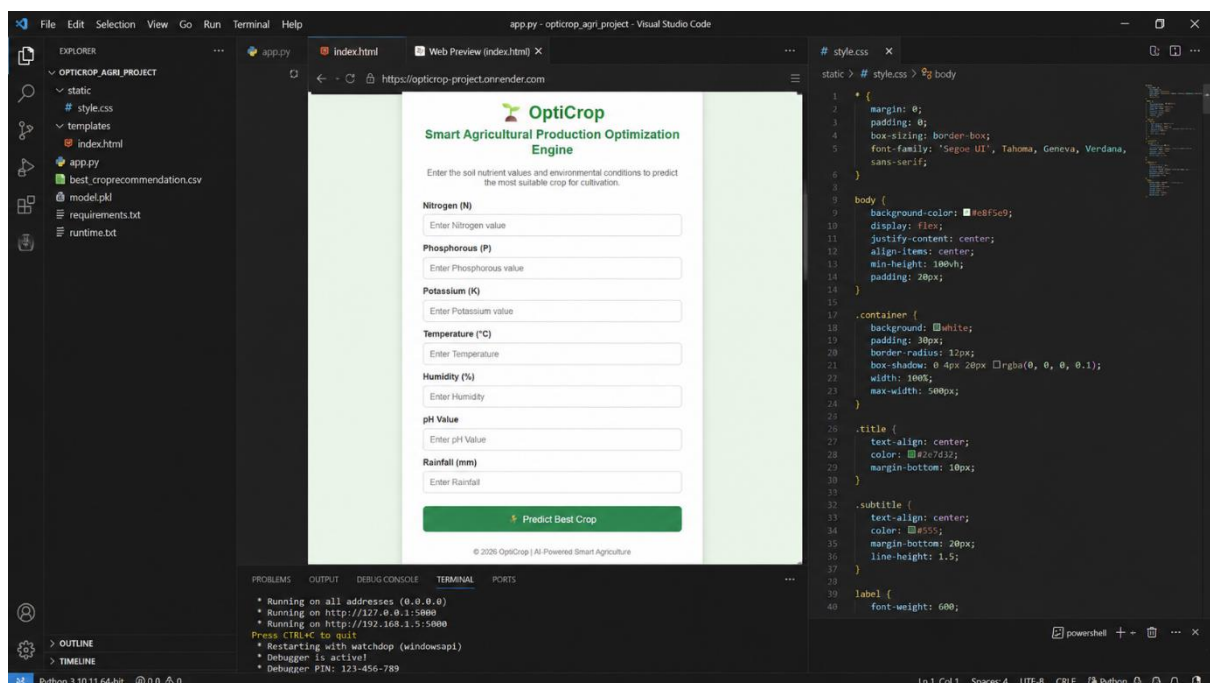
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1. Introduction

Agriculture plays a significant role in the economy of many countries. Farmers often face difficulties in selecting the most suitable crop due to changing environmental conditions and lack of accurate recommendations. Incorrect crop selection leads to poor yield and financial losses.

The OptiCrop project is designed to solve this problem using Machine Learning. The application analyzes soil nutrients and climatic conditions to recommend the most suitable crop.

The project combines Artificial Intelligence, Data Science, Machine Learning, and Web Development to provide an easy-to-use prediction system.



2. Project Overview

OptiCrop is a Machine Learning based crop recommendation system developed using Logistic Regression.

The application accepts environmental parameters such as Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall from the user.

After processing these values, the trained machine learning model predicts the most suitable crop for cultivation.

The complete system is developed using Python, Flask, HTML, CSS, and deployed on Render Cloud Platform.

3. Problem Statement

Farmers generally rely on traditional knowledge and experience while selecting crops.

This approach often leads to:

- Low crop productivity
- Wrong crop selection
- Financial loss
- Inefficient resource utilization

The project aims to provide intelligent crop recommendations using machine learning techniques.

4. Objectives

The major objectives of the project are:

- Develop an intelligent crop recommendation system.
- Analyze soil nutrients.
- Predict suitable crops using Machine Learning.
- Improve agricultural productivity.
- Provide a simple web application for users.
- Deploy the application online for public access.

5. Scope of the Project

The project can be used by:

- Farmers
- Agricultural Students
- Researchers

- Agriculture Officers
- Educational Institutions

Future integration with IoT sensors and weather APIs can make the system even more intelligent.

6. Technologies Used

Programming Language

- Python

Machine Learning

- Scikit-learn
- Logistic Regression

Data Processing

- Pandas
- NumPy

Visualization

- Matplotlib

Web Framework

- Flask

Frontend

- HTML
- CSS

Deployment

- Render

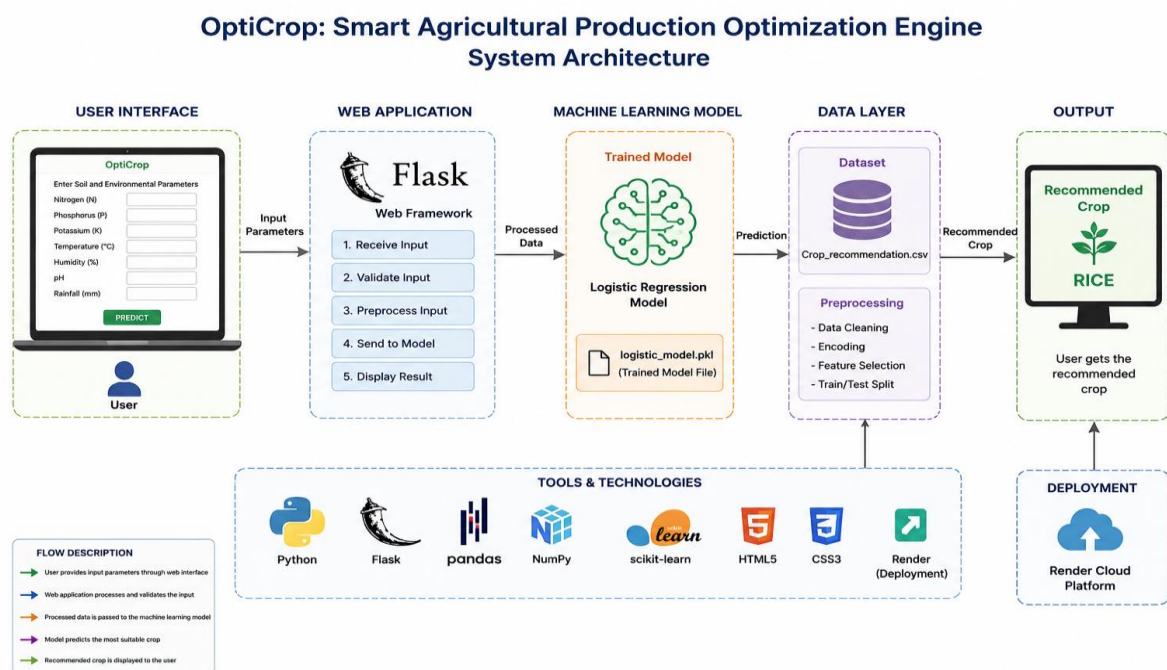
Version Control

- Git
- GitHub

7. System Architecture

The complete architecture consists of the following modules:

- User Input
- Data Processing
- Machine Learning Model
- Prediction
- Flask Backend
- HTML Interface
- Result Display



8. Project Workflow

The workflow of the project is as follows:

Problem Identification



Data Collection



Data Preprocessing



Feature Engineering



Model Training



Model Evaluation



Flask Application Development



Testing



Deployment



User Prediction

9. Dataset Description

The dataset used is Crop Recommendation Dataset.

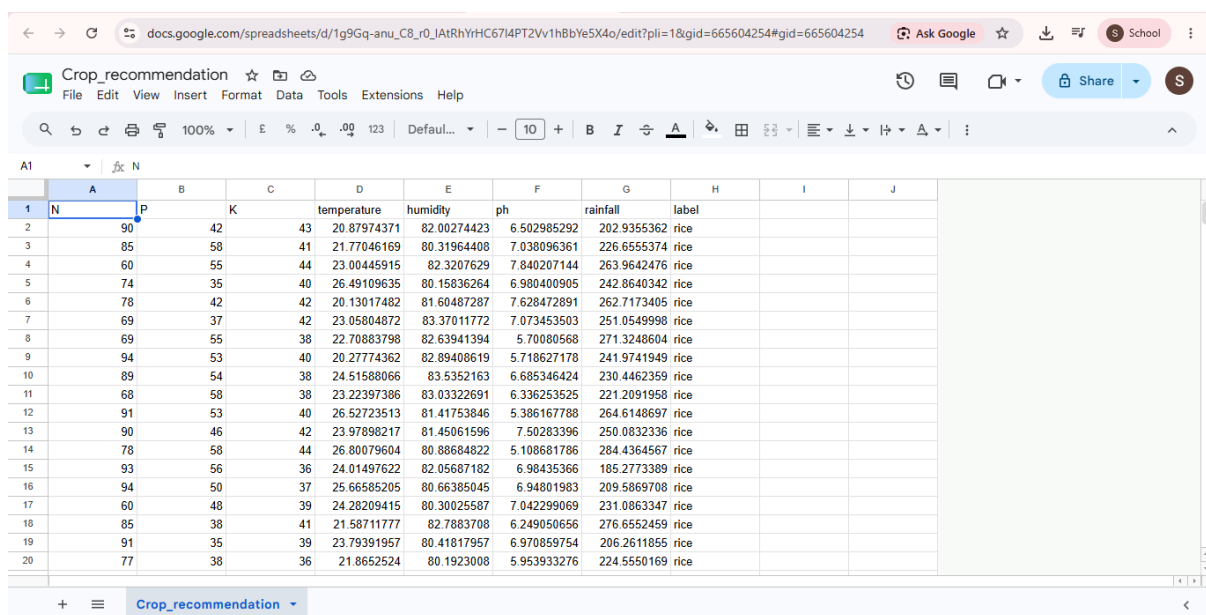
Input Features:

- Nitrogen
- Phosphorus
- Potassium
- Temperature
- Humidity
- pH
- Rainfall

Output:

- Recommended Crop

The dataset contains multiple crop categories suitable for different environmental conditions.



	A	B	C	D	E	F	G	H	I	J
1	N	P	K	temperature	humidity	ph	rainfall	label		
2	90	42	43	20.87974371	82.00274423	6.502985292	202.9355362	rice		
3	85	58	41	21.77046169	80.31964408	7.038096361	226.6555374	rice		
4	60	55	44	23.00445915	82.3207629	7.840207144	263.9642476	rice		
5	74	35	40	26.49109635	80.15836264	6.98040905	242.8640342	rice		
6	78	42	42	20.13017482	81.60487287	7.628472891	262.7173405	rice		
7	69	37	42	23.05804872	83.37011772	7.073453503	251.0549998	rice		
8	69	55	38	22.70883798	82.63941394	5.70080568	271.3248604	rice		
9	94	53	40	20.27774362	82.89408619	5.718627178	241.9741949	rice		
10	89	54	38	24.51588066	83.5352163	6.685346424	230.4462359	rice		
11	68	58	38	23.22397386	83.03322691	6.336253525	221.2091958	rice		
12	91	53	40	26.52723513	81.41753846	5.386167788	264.6148697	rice		
13	90	46	42	23.97898217	81.45061596	7.50283396	250.0832336	rice		
14	78	58	44	26.80079604	80.88684822	5.108681786	284.4364567	rice		
15	93	56	36	24.01497622	82.05687182	6.98435366	185.2773389	rice		
16	94	50	37	25.66585205	80.66385045	6.94801983	209.5869708	rice		
17	60	48	39	24.28209415	80.30025587	7.042299069	231.0863347	rice		
18	85	38	41	21.58711777	82.7883708	6.249050656	276.6552459	rice		
19	91	35	39	23.79391957	80.41817957	6.970859754	206.2611855	rice		
20	77	38	36	21.8652524	80.1923008	5.953933276	224.5550169	rice		

Crop Recommendation Dataset (CSV)

- <https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>

10. Data Preprocessing

Several preprocessing techniques were applied:

- Missing value verification
- Duplicate removal
- Feature selection
- Label Encoding
- Data cleaning
- Data validation

The processed dataset was then used for training.

11. Model Development

The machine learning model was developed using Logistic Regression.

Development steps:

- Import dataset
- Split training and testing data
- Train Logistic Regression model
- Save model using Pickle
- Load model inside Flask

The trained model is stored as:

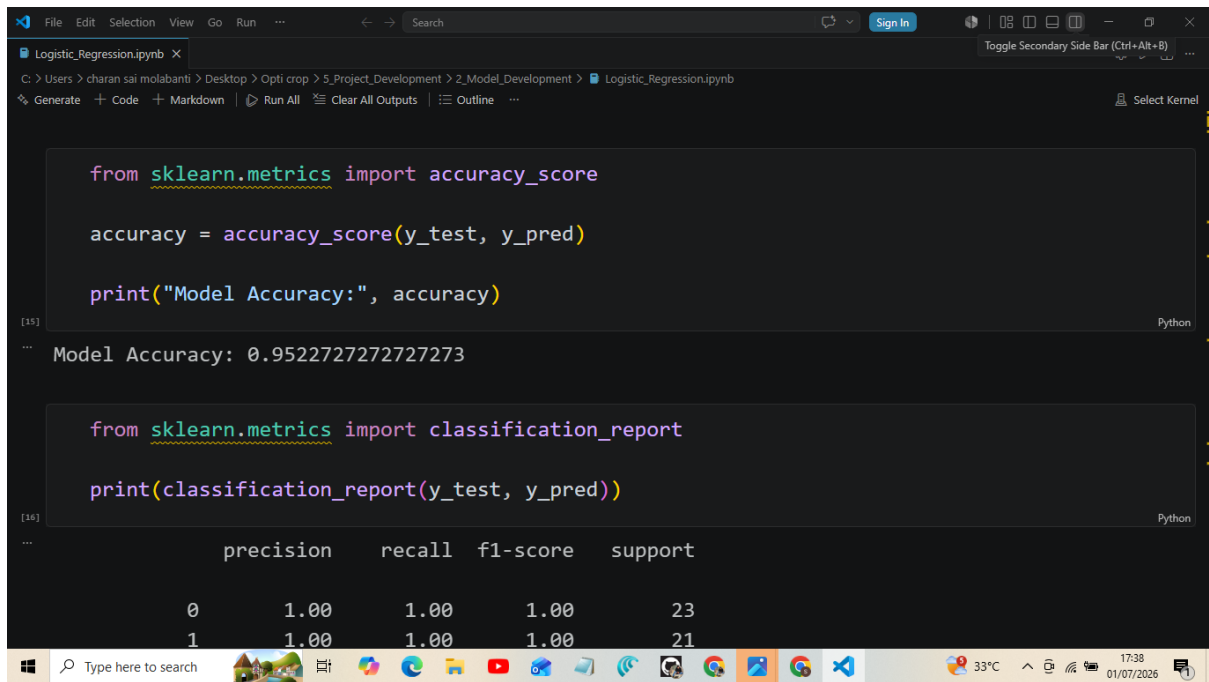
logistic_model.pkl

12. Model Evaluation

Evaluation metrics used:

- Accuracy Score
- Classification Report
- Confusion Matrix

The trained model achieved high prediction accuracy for crop recommendation.



The screenshot shows a Jupyter Notebook interface with two code cells. The first cell calculates the model accuracy using `sklearn.metrics.accuracy_score` and prints the result: "Model Accuracy: 0.9522727272727273". The second cell calculates the classification report using `sklearn.metrics.classification_report` and prints the following table:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	23
1	1.00	1.00	1.00	21

The notebook interface includes a menu bar (File, Edit, Selection, View, Go, Run), a search bar, and a status bar at the bottom showing system information like temperature (33°C) and date (01/07/2026).

13. Web Application Development

Frontend:

- HTML
- CSS

Backend:

- Flask

Features:

- User-friendly interface
- Input validation
- Instant prediction
- Responsive design

The Flask application connects the trained model with the web interface.

14. Deployment

The application was deployed using Render.

Deployment Steps:

- Push project to GitHub
- Connect GitHub Repository
- Configure Root Directory
- Install Dependencies
- Deploy Flask Application

Live Project:

<https://opticrop-project.onrender.com>

Repository:

https://github.com/Swapna4321/opticrop_agri_project

15. Testing

Testing performed:

Functional Testing

Verified prediction functionality.

Input Validation Testing

Checked valid and invalid user inputs.

UI Testing

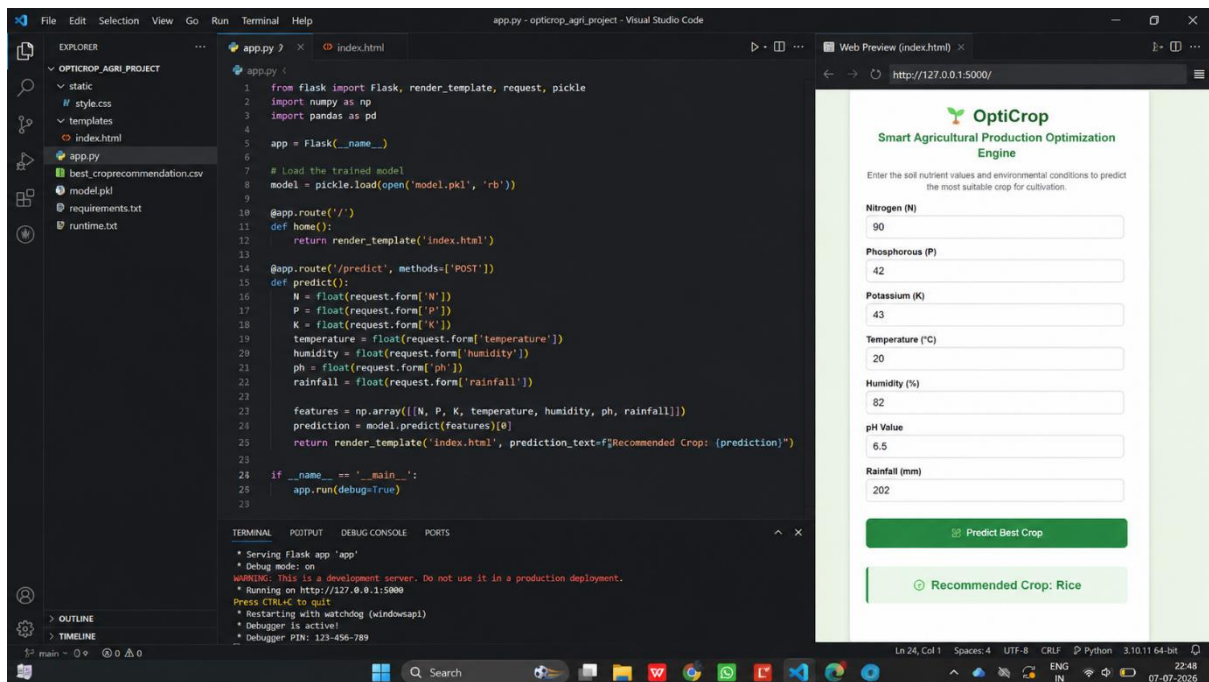
Verified interface responsiveness.

Deployment Testing

Checked cloud deployment accessibility.

Performance Testing

Verified prediction response time.



16. Results

The project successfully predicts the most suitable crop based on environmental conditions.

Results indicate that Machine Learning can effectively support agricultural decision-making.

The deployed application provides accurate predictions through an interactive web interface.

17. Advantages

- Easy to use
- Accurate recommendations
- Time-saving
- Supports farmers

- Online accessibility
- Machine Learning based
- Scalable architecture

18. Limitations

- Uses only available dataset.
- No real-time weather integration.
- No GPS-based recommendation.
- Limited to predefined crop categories.

19. Future Enhancements

Future improvements include:

- Weather API integration
- IoT sensor connectivity
- Deep Learning models
- Fertilizer recommendation
- Disease prediction
- Mobile application
- Multi-language support
- Farmer authentication system
- Cloud database integration

20. Conclusion

The OptiCrop project demonstrates how Artificial Intelligence and Machine Learning can improve modern agriculture by providing intelligent crop recommendations.

The developed Flask application successfully predicts suitable crops based on soil nutrients and climatic conditions. The project integrates data preprocessing, machine learning, model deployment, and web technologies into a single user-friendly application. It has the potential to assist farmers in making informed decisions, increasing agricultural productivity, and reducing crop selection errors. Future enhancements can further improve the system by incorporating real-time environmental data and advanced predictive models.

21. References

1. Python Documentation — <https://docs.python.org/>
2. Scikit-learn Documentation — <https://scikit-learn.org/>
3. Flask Documentation — <https://flask.palletsprojects.com/>
4. Pandas Documentation — <https://pandas.pydata.org/>
5. NumPy Documentation — <https://numpy.org/>
6. Render Cloud Platform — <https://render.com/>
7. GitHub Documentation — <https://docs.github.com/>